

PalmSource®

Conference and Expo



Programming Tips for Symbol's Rugged Palm OS® Devices

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Introduction of the Symbol SDK with tips for developing scanning and wireless applications.



Agenda…

- Introducing Symbol Technologies
- Symbol SDK - Overview
- Barcode Scanning
- Wireless Networking
- Useful Resources

Symbol Technologies, Inc.

- Technologies:
  - Mobile Computing
  - Barcode Scanning
  - Wireless Networking
- Vertical markets:
  - Education
  - Health Care
  - Hospitality
  - Manufacturing
  - Retail
  - Transportation



Symbol's Palm OS® Devices



- SPT 15xx
  - barcode-enabled Palm III devices
- SPT 17xx
  - Ruggedized, barcode-enabled, wireless LAN and WAN options
- SPT 18xx
  - Like SPT17xx – faster processor
- CSM 150
  - Barcode scanning Springboard module
- Accessories
  - Multi-slot cradles, keyboards, protective covers, vehicle cradles, magnetic card readers, and external battery chargers

Symbol University

- Award winning corporate university
- Official Training Center for Palm OS Developer classes:
  - 5-day Palm OS Essentials
  - 3-day Conduit Development
  - 2-day Communications
- Certified Palm OS Trainers, based in the U.S.A. and Europe
- http://edu.symbol.com



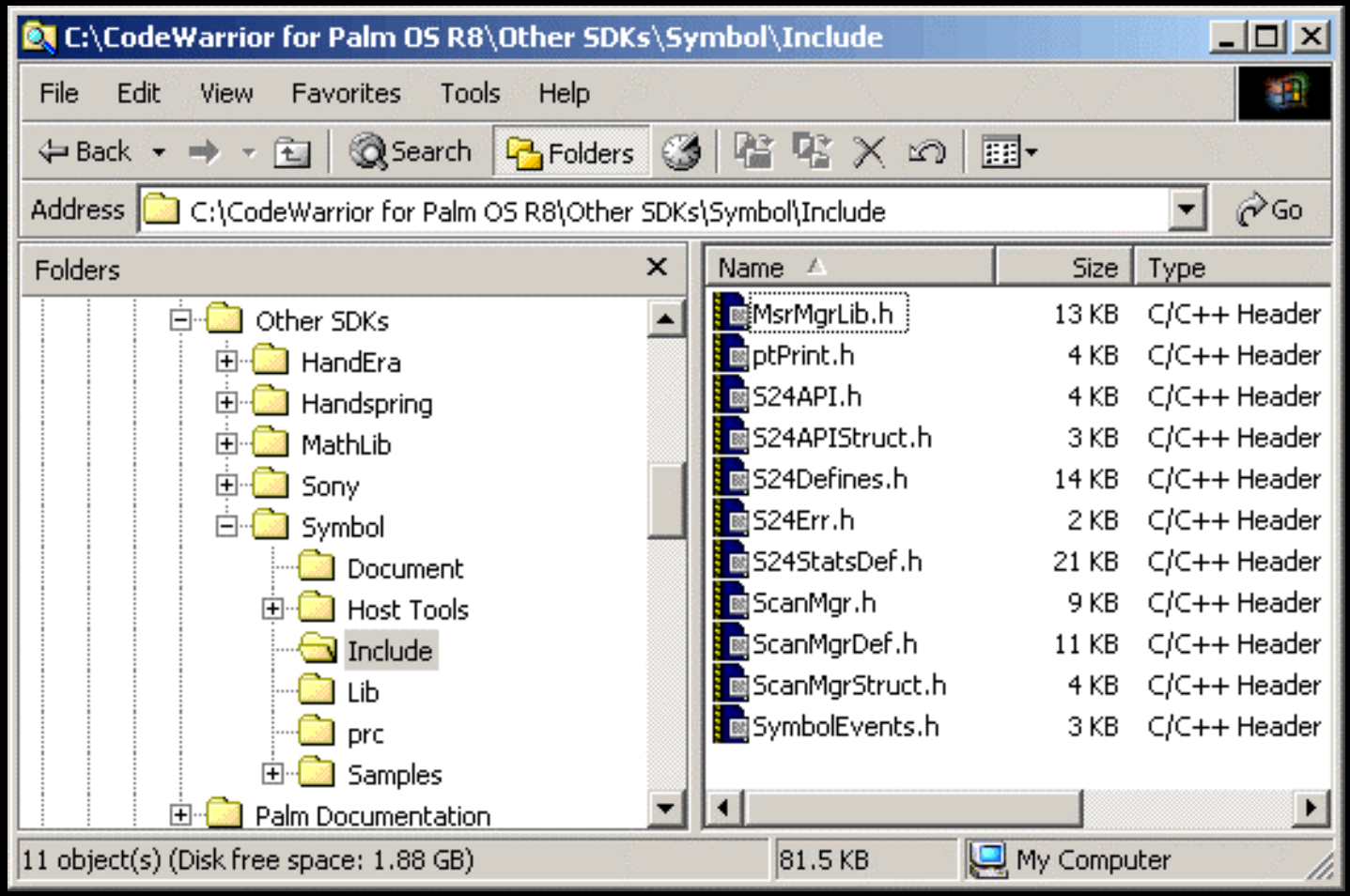


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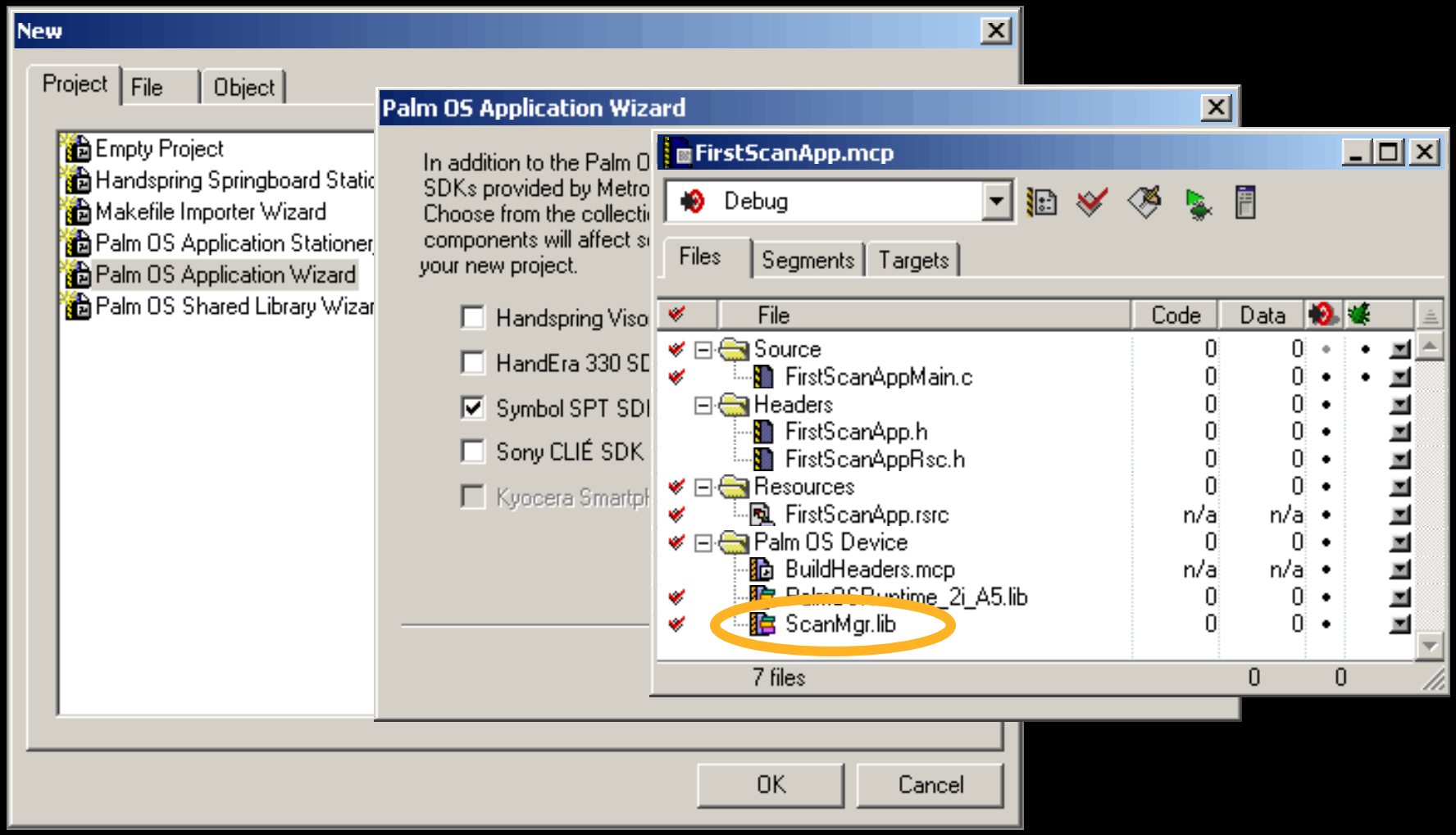
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Symbol SDK - Overview



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CodeWarrior R8 – Application Wizard





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Barcode Scanning – Step by Step

1. Include necessary header files
2. Open scan decoder
3. Set scanning parameters
4. Enable scanning
5. Handle scan events
6. Disable scanning
7. Close scan decoder

1. Include the necessary header files

```
#include <PalmOS.h>

/* Symbol SDK */
#include "ScanMgrDef.h"          /* constants */
#include "ScanMgrStruct.h"       /* structures */
#include "ScanMgr.h"             /* API functions */

#include "YourProject.h"
#include "YourProjectRsc.h"
```

2. Open Scan Decoder

```
static Err AppStart(void)
{
  /* check if we're running on Symbol hardware */
  if (ScanIsPalmSymbolUnit()) {
    Err error = ScanOpenDecoder(); /* open lib */
    if (!error) {
      bOpenDecoderOK = true;   /* set global */
      ScanCmdScanEnable();     /* Enable scanning */
    }
  }
}
```

Tip: When you have more than one form, only enable scanning on forms that actually provide scanning functionality.

3. Set Scanning Parameters

```
static Err AppStart(void)
{
    if (!error) {
       bOpenDecoderOK = true;                    /* set global */
       ScanCmdParamDefaults(); /* Reset to defaults */
       ScanSetTriggeringModes(LEVEL);    /* set mode */
       /* Enable only symbologies we'll use */
       DisableAllBarcodes();
       ScanSetBarcodeEnabled(barUPCA, true);
       ScanSetBarcodeEnabled(barCode128, true);
       ScanCmdSendParams(No_Beep); /* Send parameters
                                      to decoder */
    }
  }
```

LEVEL
PULSE
HOST

← you write this function

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4. Enable Scanning

```
static void ScanFormInit(FormType *frmP)
{
  /* check if we are on Symbol hardware */
  if (bOpenDecoderOK) {
      ScanCmdScanEnable();     /* Enable scanning */
    }
}
```

5. Handle Scan Events

- Scanner Events:
  - **`scanDecodeEvent`**
    - Length of scanned data
    - Type of barcode
    - Status (true if scan was successful)
    - Data from the barcode
  - **`scanBatteryErrorEvent`**
    - Battery is too low for scanning
  - **`scanTriggerEvent`**
    - A scan attempt was initiated – soft or hard trigger
    - You usually don't trap this event!

5. Handle Scan Events (cont'd)

```
static Boolean ScanFormHandleEvent(EventType * eventP)
{
  case scanDecodeEvent: {
    MESSAGE decodeDataMsg;
    int status = ScanGetDecodedData(&decodeDataMsg);

    if( status == STATUS_OK ) {
      if(decodeDataMsg.type != 0) {
        StrCopy(gTracking, (Char *)decodeDataMsg.data);
      }
    }
    handled = true;
}
```

5. Handle Scan Events (cont'd)

```
static Boolean ScanFormHandleEvent(EventType * eventP)
{
  case scanBatteryErrorEvent:
       FrmAlert(MyBatteryErrorAlert);
       handled = true;
       break;
}
```

5. Handle Scan Events – 2D Barcodes

```
typedef struct
{
  enum events eType;
  Boolean penDown;
  Int16 screenX;
  Int16 screenY;
  union scanData {
    struct scanGen {
      UInt16 data1; //Bit 0 is the flag to tell app
                    //if decode data has extended data.
      UInt16 data2; //Length of the decoded data.
      UInt16 data3;
      ...
```



5. Handle Scan Events – 2D Barcodes

```
static Boolean ScanFormHandleEvent(EventType * eventP)
{
  case scanDecodeEvent:
    extendedDataFlag=((ScanEventPtr)event)
                                    ->scanData.scanGen.data1;
    extendedDataLength=(Int16)(((ScanEventPtr)event)
                                    ->scanData.scanGen.data2);
    if (extendedDataFlag & EXTENDED_DATA_FLAG) {
      extend = true;
    } else {
      extend = false;
    }
    OnDecoderData(extend);
    handled = true;
    break;
}
```

5. Handle Scan Events – 2D Barcodes

```
Boolean OnDecoderData(Boolean extend)
{
  if (extend){
    hExtendedData = MemHandleNew (extendedDataLength);
    pExtendedData = MemHandleLock (hExtendedData);
    status = ScanGetExtendedDecodedData(extendedDataLength,
             &extendedDataType, pExtendedData);
  } else {
    status = ScanGetDecodedData(&decodeDataMsg);
  }
}
```

6. Disable Scanning

```
static void ScanFormDeInit(FormType *frmP)
{
  /* check if we are on Symbol hardware */
  if (bOpenDecoderOK) {
      ScanCmdScanDisable();   /* Disable scanning */
    }
}
```

7. Close Scan Decoder

```
static Err AppStop(void)
{

 /* check if we opened decoder earlier */
 if (bOpenDecoderOK) {
    ScanCloseDecoder(); /* close decoder lib */
    }
}
```

Barcode Scanning – Tips

- Only enable scanner on forms you provide scan functionality.
- Only enable those barcode types you need and explicitly disable all others to gain scan performance.
- Provide alternative means to input data.
- Trap `scanBatteryErrorEvent` and display alert with instructions.
- Consider deactivating system buttons.

Disabling Hardware Buttons in EventLoop

```
do {
  EvtGetEvent(&event, evtWaitForever);

  // Intercept System Buttons
  if (! InterceptSystemButtonsHandleEvent(&event))
    if (! SysHandleEvent(&event))
      if (! MenuHandleEvent(0, &event, &error))
        if (! AppHandleEvent(&event))
            FrmDispatchEvent(&event);

  if ((event.eType == appStopEvent) && (isNewRecord))
       event.eType = nilEvent;

  } while (event.eType != appStopEvent);
```



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Wireless Networking Tips

SPT 17XX / 18XX

WLAN (Spectrum24)

- 1742–2D, 1842 802.11 2 Mbps
- 1746–2D, 1846 802.11b 11 Mbps

- WWAN
  - 1833 CDPD
  - 1834 GSM
  - 1837 GSM + GPRS



Wireless Networking Tips

SPT 17XX / 18XX

- 1700
  - DragonBall 68328 16 MHz
- 1800
  - DragonBall VZ 33 MHz Processor
- OS version 3.52
  - 4.1 planned

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Wireless Networking Tips

Latest SDK Version 3.10

- Available from Symbol:
  - http://devzone.symbol.com
    - Developer Downloads
    - Palm OS®
- Also available
  - ROM Builder
  - ROM Images

Wireless Networking Tips

Documentation Folder

- CDPDapi.pdf
  - SPT1733 Developer's Guide
- S24api.pdf
  - SPT174X Spectrum24® Driver Extensions Library Developer's Guide
- WWanSupGuide.pdf
  - WWAN Supplement Guide
    - SPT1734 with GSM wireless option

Wireless Networking Tips

Include Folder

- S24API.h
- S24APIStruct.h
- S24Defines.h
- S24Err.h
- S24StatsDef.h

Lib Folder

spt184x.lib

Wireless Networking Tips

S24 Samples Folder - spt1740_ReadMe.htm

- Client (CodeWarrior projects)
  - CaDmv
    - MSR + S24
  - Echo
    - Sockets API test routines
    - Unix Network Programming by Richard Stevens
  - FindMdb
    - Scan a barcode, send to server, receive description
- Server
  - Host applications

Wireless Networking Tips

- Don’t use Berkeley socket calls if global variables aren’t available
  - Berkeley Sockets
    - `Sendto(socket, buf, buflen, flags, &addr, sizeof(addr))`
  - PalmOS NetLib
    - `NetlibSend(AppNetRefnum, socket, buf, buflen, flags, &addr, sizeof(addr), AppNetTimeout, &errno)`
  - Conversion macros in `sys_socket.h`
    - Useful for porting existing code

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Wireless Networking Tips

- **`NetLib`** redirection of POSE is good for development
  - But test on actual device
  - Redirection doesn't use the OS TCP stack
  - Memory issues (32KB for TCP stack)
  - Turn redirection off (use PPP server)
- TCP/IP may already be running when an application is launched
  - Less memory available
  - **`NetLibClose(AppNetRefnum, immediate)`**

Wireless Networking Tips

- Read and write routines may not transfer the requested amount of data
  - **`NetUReadN`**
  - **`NetUWriteN`**
- Use **`NetLibSelect`** as an alternative to polling
  - UI remains active
  - Socket ready for reading / writing
  - User input event

Wireless Networking Tips

- Scanner priority over RF
  - No transmit State (Tx Inhibit)
  - Don’t allow the user to perform rapid scans
    - Inhibit RF transmissions for prolonged periods.
    - Loss of association with the AP
- Napping (Stage 1)
  - Auto-off timer expires
  - Radio enters PSP-10 mode until stage1-timer expires
  - Default setting 10 mins (`S24SetPreference`)

Wireless Networking Tips

- Include **`spt184x.lib`** in your project files
  - Insulates the application from detail of using shared libraries
  - Aids porting existing code
- **`S24GetAssociationStatus`**
  - Out of range, roaming, radio network down
  - After 5 seconds inform user
  - Sit in loop waiting for reassociation
    - Stop sending data out through NetLib

Wireless Networking Tips

- Useful Spectrum24 functions
  - **S24GetAPTable**
  - **S24GetBeaconParams**
  - **S24GetCountryCode**
  - **S24GetInfo**
    - **S24Info structure (S24APIStruct.h)**
  - **S24GetPreference**
    - network preferences **(S24Defines.h)**
  - **S24GetPreferredBSSID**
  - **S24GetAdapterESSID**



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Useful Resources

- SDK Documentation
- SDK Example Projects
- Symbol Developer Zone
  - www.symbol.com/palm
- Symbol Developer Forum
  - www.palmos.com/dev/support/forums
- Barcoding Information
  - www.adams1.com/pub/russadam/barcode1.html



Questions

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